High Performance Data Analytics and Visualisation

Assignment 1 – Ushakov Andrii

1. Characterization of the Problem, Data, and User Tasks

The goal of this visualization is to help a user explore the US Communities and Crime dataset in order to compare cities/communities and reason about where it may be safer or more suitable to live. The dataset contains many normalized attributes for each community, including state, communityname, population, medIncome, and ViolentCrimesPerPop.

The main user tasks are to identify communities with high or low violent crime, compare them across states, and understand how crime relates to population. To support these tasks, the interface combines a scatterplot and a hierarchical view. The scatterplot helps compare two quantitative attributes, while the hierarchy groups communities by state and shows the distribution of communities inside each state.

Figure 1 shows the initial state of the application. Figure 2 shows selecting communities using the scatterplot/treemap interaction. Figure 3 shows the alternative circle-pack layout. Figure 4 shows the hover interaction, where a user can inspect details for a specific community.

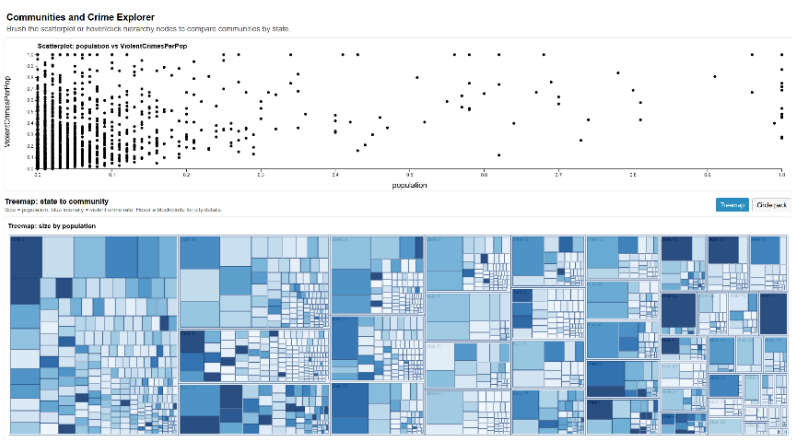

Figure 1

Figure 2

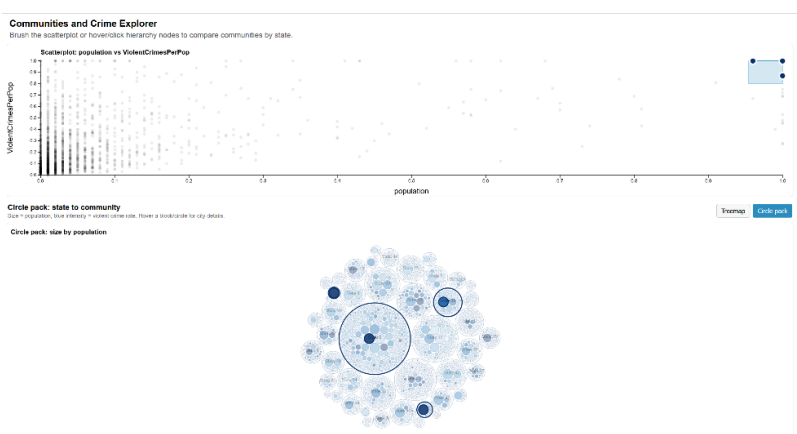

Figure 3

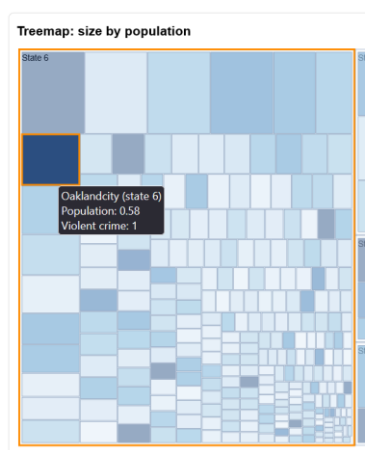

Figure 4

2. Design Rationale

The application is based on Tutorial 5, which provided the React, Redux, and D3 design pattern used as the foundation. I reused the idea of separating React container components from D3 rendering classes. React and Redux manage the application state and interactions, while the D3 classes render and update the visualizations.

The scatterplot uses population on the x-axis and ViolentCrimesPerPop on the y-axis. Each dot represents one community. This design allows the user to quickly see whether high-crime communities are concentrated in small or large populations. A 2D brush interaction lets the user select multiple communities at once.

The hierarchy uses two levels: state and communityname. The main hierarchy layout is a treemap, where each state contains its communities. The size of each block represents population, and the blue color intensity represents ViolentCrimesPerPop. This makes the treemap useful for comparing both population and crime intensity in a compact view. Details such as community name, state, population, and crime value are shown on hover instead of displaying too many labels permanently.

3. Interaction and Layout Comparison

The visualizations are coordinated. When the user brushes points in the scatterplot, the corresponding communities are highlighted in the hierarchy. When the user clicks or hovers over a hierarchy block, the corresponding point is highlighted in the scatterplot. To make the selected points easier to find, I changed the scatterplot highlighting so selected dots become larger, darker blue, and outlined, while non-selected dots fade.

I tested two D3 hierarchy layouts: treemap and circle pack. The circle-pack layout is useful because it clearly shows containment using circles, and it gives an alternative view of the state-community hierarchy. However, for this dataset it becomes harder to read because there are many communities, so many circles become very small and labels overlap or disappear. The treemap is more space-efficient and makes better use of the available rectangular screen area. For this reason, my preferred layout is the treemap. The circle-pack layout is still available through a switch so the user can compare both layouts, but the treemap is the main layout I would choose for the final analysis.

4. Declaration of Generative AI Use

I used Codex as a coding assistant during the development of this project. The base of the project was Tutorial 5, and Codex helped me understand how to extend this base into the assignment structure. I asked Codex for guidance on how the visualizations could be structured and how the assignment requirements could be mapped to React, Redux, and D3 components.

The first structure suggested by Codex showed multiple views side by side, but I rejected that version because it made the interface too crowded. I then corrected the design direction and

decided to use a switch between the treemap and circle-pack hierarchy layouts in the same bottom section. This made the layout cleaner and easier to compare. I also noticed that the selected scatterplot points were hard to see, so I changed the highlighting idea to make selected dots larger and more visible.

I accepted the general React/Redux/D3 implementation support from Codex, but I adapted the visual design decisions based on what looked clearer in the browser. The final choices, especially preferring the treemap layout, reducing labels, using hover details, and improving highlight visibility, were made after inspecting the visual output myself. The prompts and Codex answers are provided in a separate document as required.